

AI Backlog - FAMU Career Assistant

Team Members: Calvin Mitchell, Danielle Devose, Jeremy Metellus, Nathaniel Washington

Introduction

This report documents the use of AI tools during the development of the FAMU Career Assistant project. It highlights how AI was integrated into the project workflow, from backlog creation to user story drafting and sprint planning. The goal is to demonstrate transparency, ethical use, and the contribution of AI to our project.

AI Tools Used

- **ChatGPT** – used for user story drafting, acceptance criteria writing
- **Google Gemini** - The Gemini model was used to brainstorm, structure, and refine the core components of the career assistant program.
- **Junie** - has a systematic, multi-stage process that simulates a human's logical approach to problem-solving. Instead of giving an immediate, surface-level response, she breaks down a request into a detailed, actionable plan

Danielle

Danielle's Prompt History

Prompt: help build the dashboard and design flow for our project

Answer:

PLAN:

- **Dashboard Inspiration:** Gathered reference screenshots and design photos to visualize how the dashboard should look. Used these as a foundation to guide layout, navigation, and component structure.
- **Save Page Feature:** Drafted the Save page design, ensuring it matched the style and flow of the dashboard. This included identifying which components needed user interaction (save buttons, applied jobs panel, etc.).
- **Architecture File:** Created the architecture.md file to document the system structure. This described how the frontend (React/Vite) and backend (Node/Express with MongoDB) connect, what services are in play, and how requests flow between them.

- **Feedback Iteration:** Adjusted the design and documentation after reviewing the reference images and team input, refining the page flow for a more user-friendly experience.
- **Deliverables:**
 - Reference-based dashboard design outline
 - Save page draft and interactions
 - architecture.md documenting the system flow

Evidence of AI Use

Save Page evidence: <https://chatgpt.com/share/68d1ac91-a328-8013-9208-3e90570d42df>

Architecture evidence: <https://chatgpt.com/share/68d1aae2-83e0-8013-9557-1abbbbdee5d9f>

Output Evaluation

Technical guidance

- Suggested how to translate image references into concrete UI components (e.g., panels, buttons, and layouts).
- Provided structure for the architecture.md file to keep it consistent with the codebase and project standards.

Troubleshooting support

- Highlighted potential pitfalls in frontend/backend communication and how to document them clearly.
- Advised on iterating designs using visual references while staying aligned with technical feasibility.

CALVIN

Prompt History

Junie's Prompt History

Prompt: add the necessary information in the readme to properly demo this program **Answer:**
PLAN:

1. **Inspect Key Files:** Examine frontend files like `src/main.jsx` and `vite.config.js` to grasp the application's entry point, routing, and how it handles ports and proxies. This helps determine how a user would access the application and what dependencies exist.
2. **Review Server Endpoints:** Analyze the backend API in `server/index.js` to get a summary of all the features available for a demo. This includes critical functions such as user registration, login, questionnaire completion, and health status checks. Knowing the endpoints is crucial for explaining the program's functionality.
3. **Update `docs/README.md`:** Create detailed, step-by-step instructions for demonstrating the program. This section will include:
 - **Prerequisites:** List what a user needs to install.
 - **Quick Start:** Provide a clear command-line interface (CLI) method for running the program without an IDE.
 - **Environment Setup:** Explain how to configure the `.env` file with essential variables like `MONGODB_URI`, `PORT`, and any optional keys for logging or AI services.
 - **Running the Server:** Describe the commands to run the development server and explain the expected URLs. Clarify how the Vite proxy connects the frontend to the backend.
 - **Demo Flow:** Walk through a typical user journey with expected outcomes at each step.
 - **API Testing:** Show how to test the API endpoints quickly using `curl` commands.
 - **Troubleshooting:** Address common issues like port conflicts, database connection problems, and how to check logs and health endpoints.
4. **Validate Instructions:** Cross-check the instructions against the actual code to ensure all file paths, script names, and port numbers are correct and consistent.
5. **Save Changes:** Finalize and save the updated `docs/README.md` file.
6. **Provide Summary:** Conclude with a concise summary of the completed work.

Evidence of AI Use

```
add the nessisary information in the readme to properly demo
this program
```

Plan

- 1. Inspect key frontend files (src/main.jsx, vite.config.js)
✓ to understand app entry, routes, ports/proxy, and any environment usage.
- 2. Review server endpoints in server/index.js to summarize
✓ available features needed for a demo (register, login, questionnaire, health).
- 3. Update docs/README.md with comprehensive demo instructions:
 - ✓ - Prerequisites and quick start via npm CLI (without IDE).
 - ✓ - Environment setup (.env) including MONGODB_URI, PORT, optional LOG_FILE, and optional AI keys.
 - ✓ - How to run dev server and expected URLs; explain Vite proxy to backend.
 - ✓ - Step-by-step demo flow with expected behavior.
 - ✓ - API endpoints quick tests via curl.
 - ✓ - Troubleshooting (ports, MongoDB connection, logs, health checks).
 - ✓ - Retain optional IntelliJ configuration.
- ✓ 4. Validate instructions for consistency (ports, scripts, paths) by cross-checking code.
- ✓ 5. Save changes to docs/README.md.
- ✓ 6. Provide final summary and submit.
-

Changed Files

README.md +118 -18

The README.md was extensively updated with detailed instructions for properly demonstrating the program. The documentation includes prerequisites, environment setup, running the app, a step-by-step demo walkthrough, API test examples, IntelliJ configuration, and troubleshooting advice. No code changes were made; only documentation was enhanced to enable a comprehensive demo.

Output Evaluation

Technical guidance

- Highlighted common configuration elements (e.g., .env usage, API keys, port settings) and safe handling practices.
- Provided suggestions for organizing files and improving readability of project docs.

Troubleshooting support

- Offered likely causes and fixes for typical local dev issues (e.g., ports, environment variables, config mismatches).
- Proposed step-by-step checks to validate assumptions before deeper debugging.

JEREMY

Prompt History

Prompt: Help build an outline for the job listing dashboard on the website

TASK: To create a job posting dashboard for the website and find a fun way to integrate different needed items within the website like ai and filter options

Plan

Add a filter options on the side of the search to allow the user to filter through the jobs they want

Add color to the parts of the website

Add a portion of the website where we can integrate a database when necessary for the operation of the website.

Output evaluation

After many queries, I think I was able to get a product that was sufficient. I decided to take some of the code it gave me and edit it ,and then i added it to the

Evidence of AI Use

<https://chatgpt.com/share/68d1e688-0268-8007-a170-0a5901a38a2b>

NATE

Prompt History

CHATGPT Prompt History

Prompt: help to generate example queries for my Mongoose models.

Task: Create sample MongoDB queries for testing my database models (User, AdvancedQA, Results, etc.)

PLAN:

- Generate queries with CRUD operations (create, read, update, delete).
- Included test data insertions (e.g., creating a user, adding a job posting).
- Provided functions to query and update data for demo purposes.
- Generate the file queries.js

Evidence of AI Use

```
// ----- USERS -----
export async function createUser(email, passwordHash, userTag) : Promise<any> { no usages
  return await User.create({ email, passwordHash, userTag });
}

export async function getUserByEmail(email) : Promise<any> { Show usages
  return await User.findOne({ email });
}

export async function getUserById(userId) : Promise<any> { no usages
  return await User.findById(userId);
}

export async function updateUserTag(userId, newTag) : Promise<any> { no usages
  return await User.updateOne({ _id: userId }, { $set: { userTag: newTag } });
}

export async function deleteUser(userId) : Promise<any> { no usages
  return await User.deleteOne({ _id: userId });
}
```

```
// ----- ADVANCED QA -----
export async function addAdvancedQA(userId, genericQ, genericA, aiQ, aiA, profile) : Promise<any> {
  return await AdvancedQA.create({
    userId,
    genericQuestions: genericQ,
    genericAnswers: genericA,
    aiQuestions: aiQ,
    aiAnswers: aiA,
    profileSnapshot: profile,
  });
}

export async function getAdvancedQAByUser(userId) : Promise<any> { Show usages
  return await AdvancedQA.find({ userId });
}

export async function updateAIAnswers(userId, aiAnswers) : Promise<any> { no usages
  return await AdvancedQA.updateOne({ userId }, { $set: { aiAnswers } });
}

export async function deleteAdvancedQA(userId) : Promise<any> { no usages
  return await AdvancedQA.deleteOne({ userId });
}
```

Output Evaluation

- Saved time writing repetitive test queries manually; gave me a working file I could run to confirm my models and MongoDB connection work.
- Without AI, I would have had to manually code and debug the queries. AI helped me focus on connecting and testing faster.

Ethical and Professional Reflection

As a team, we approached our first sprint with professionalism and accountability, ensuring that the dashboard, save page, job listings, and databases were built to meet user needs and system requirements. We focused on collaboration, code quality, and fairness so that the foundation of our system is reliable, transparent, and ready to support future growth responsibly.

Project Contribution

As a team, AI helped us brainstorm ideas, break down features into user stories, and draft acceptance criteria. It gave us a starting point that we refined together, which made the process faster, more organized, and easier to manage.

Sign-Off

We confirm this log accurately reflects the team's AI usage.

Team Members (type names):

- Danielle Devose

- Calvin Mitchell

- Jeremy Metellus

-Nathaniel Washington